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AUTOMATIC INTERPRETATION OF X-RAY IMAGES USING COMPUTER VISION

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Abstract. *Modern veterinary diagnostics requires highly accurate and efficient analysis methods. Radiography remains one of the key imaging tools; however, the interpretation of images can be subjective and time-consuming. This article examines the application of computer vision technologies for the automatic interpretation of X-ray images in animals. The principles of machine learning algorithms, their advantages—such as increased diagnostic speed and accuracy, and reduced human factor influence as well as existing limitations are analyzed. Special attention is paid to the prospects of integrating this technology into the educational process of veterinary students and its impact on the future of the profession. It is concluded that the automation of diagnostic image analysis is becoming an integral element of the development of digital veterinary medicine.*

Keywords: *computer vision, artificial intelligence, veterinary diagnostics, radiography, automatic image interpretation, machine learning, deep learning, digitalization of veterinary medicine.*

The development of veterinary medicine is inextricably linked with the introduction of innovative technologies capable of improving the efficiency and accuracy of diagnostics. Radiographic examination is a routine but highly informative method used in daily practice. However, traditional interpretation of X-ray images depends on the experience and qualification of the specialist, which can lead to diagnostic errors and delays in diagnosis. In this context, computer vision technologies based on artificial intelligence (AI) open new horizons for automating and objectifying the process of medical image analysis [2]. The purpose of this article is to analyze the current state, advantages, and challenges associated with the automatic interpretation of X-ray images in veterinary medicine, as well as to consider its educational potential.

Research in the application of AI for medical image analysis is actively conducted in both human and veterinary medicine. Foreign authors, such as J. Smith and K. Roberts, in their works demonstrate the high effectiveness of convolutional neural networks (CNNs) in detecting pathologies such as canine hip dysplasia or feline cardiomegaly, showing accuracy comparable to experienced radiologists [6]. Domestic research, for example, works by the team led by A.V. Petrov, also confirms the promise of using machine learning algorithms for screening musculoskeletal pathologies in farm animals [3]. Despite the growing number of publications, the issue of comprehensive implementation of these technologies into the clinical and educational practice of veterinary medicine in Russia remains open.

The principle of automatic interpretation systems is based on deep learning. Algorithms are trained on large datasets consisting of annotated X-ray images – where an expert has indicated the location and type of pathology. After training, the model can detect similar anomalies on new, previously unseen images [5].

The key advantages of this approach are:

1. Speed and efficiency: the algorithm can analyze an image in seconds, which is critical in emergency situations.

2. Increased accuracy: AI helps reduce the number of false negatives and false positives, acting as a «second opinion».
3. Reduced subjective factor: the system is not susceptible to fatigue or cognitive biases.
4. Screening and triage: the technology can be used for mass preventive examination of animals, highlighting images with a high probability of pathology for subsequent in-depth study by a doctor.

However, the implementation of the technology faces a number of challenges:

- Lack of quality data: training reliable models requires extensive and diverse annotated datasets, the creation of which is labor-intensive.
- The «Black Box» problem: it can be difficult to understand exactly which features the algorithm used to make a particular decision, which causes distrust among practicing doctors.
- Need for adaptation: a model trained on data from one animal species may perform poorly with another, requiring additional fine-tuning.

In this regard, an urgent task is the training of future veterinary specialists capable of working in tandem with AI. Integrating the fundamentals of computer vision and data analysis into the curricula of specialty 05.36.01 "Veterinary Medicine" will allow graduates to confidently use advanced diagnostic tools [4].

Commercial and scientific platforms already exist today that offer services for the automatic analysis of veterinary X-ray images. For the successful implementation of the technology in clinics and universities, it is necessary:

- Develop cooperation between veterinary universities, IT companies, and clinics to create open datasets.
- Include modules on working with diagnostic decision support systems in advanced training programs for practicing doctors.
- Stimulate scientific research aimed at validating the effectiveness of algorithms for various animal species and pathologies [1].

Thus, the automatic interpretation of X-ray images using computer vision represents a revolutionary direction in veterinary diagnostics. Despite the existing technical and adaptation complexities, the potential of this technology for improving the quality, speed, and accessibility of veterinary care is immense. Active work on its integration into clinical practice and educational programs is the key to training competitive specialists and the further development of veterinary science in Russia. In the near future, AI will become an integral assistant to the veterinarian, taking on routine tasks of image analysis and allowing the focus to be on comprehensive diagnosis and patient treatment.

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PROTECTION OF NEURAL NETWORKS AGAINST COMPETITIVE ATTACKS

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Abstract. *Currently, neural networks have become an important tool in various fields, including computer vision, natural language processing, and autonomous systems. However, their vulnerability to adversarial attacks, which can significantly degrade the performance of models, has attracted the attention of researchers and practitioners. This article analyzes existing methods for protecting neural networks from adversarial attacks, including defense-based training methods, regularization, and the use of defensive architectures. The article provides information on the effectiveness of these methods on different types of adversarial examples and present a comparison of their robustness. In conclusion, future research directions in this area and the importance of developing more sustainable models are discussed.*

Keywords: *neural networks, adversarial attacks, protecting, sustainable models.*

Neural networks have become the foundation of many modern technologies, but despite their impressive achievements, neural networks face an important